

# NEIL J. MARSHALL

QA and Testing Engineer | AI Agent-Driven Test Systems | Published Practitioner

Fremont, CA | (510) 556-0332 | njmarshall@gmail.com | [linkedin.com/in/njmarshall](https://www.linkedin.com/in/njmarshall) | [github.com/njmarshall](https://github.com/njmarshall)

## PROFESSIONAL SUMMARY

Senior SDET and AI Test Automation Architect with 15+ years of enterprise experience building agent-driven test systems that verify software at scale. Published author of 'The Test Nobody Wrote: How I Built an AI Agent That Hunts for Untested Risks' — a practitioner article on using AI agents for exploratory testing of running applications, directly describing what SFC wants to build. Deep expertise in LLM-powered test generation using the Anthropic Claude SDK (SFC's own tooling), DeepEval evaluation, self-healing orchestration, and AI observability. Python is my primary scripting language. SQL and Postgres data validation at 12+ years. U.S. based, fully remote capable, Bay Area in-person availability.

## PUBLISHED WORK AND RECOGNITION

### *The Test Nobody Wrote: How I Built an AI Agent That Hunts for Untested Risks*

Published LinkedIn article documenting the design and implementation of an AI agent that performs exploratory testing against running applications, identifies untested risk areas, and generates targeted test cases to close coverage gaps. This is precisely the agent-driven QA approach SFC is looking to build. The article covers prompt engineering for exploratory agents, non-deterministic output validation, and how to make agent test results reliable and actionable for engineering teams.

**Testing Spark: Best Practices.** Co-presenter, Spark Summit 2014 (now Databricks Data + AI Summit). Covered production QA strategies for 1,000-node elastic Spark clusters; attended by engineers from Netflix, Databricks, and Yahoo.

## CORE COMPETENCIES

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|-----------------------------------------------------|--------------------------------------------------------|
| ✓ AI Agent-Driven Exploratory Testing               | ✓ LLM-Powered Test Generation (Claude SDK)             |
| ✓ Automated and Agent-Driven Test Systems at Scale  | ✓ DeepEval LLM-as-Judge Output Validation              |
| ✓ Unit, Integration, Regression, and E2E Automation | ✓ Self-Healing Test Orchestration and JIRA Audit Trail |
| ✓ Python, Bash, JavaScript, TypeScript              | ✓ Postgres and SQL Data Validation (12+ yrs)           |
| ✓ CI/CD: GitHub Actions, Jenkins, CircleCI          | ✓ AI Observability and Drift Detection                 |
| ✓ Playwright, Selenium, Cypress, RestAssured        | ✓ Snapshot and Visual Regression Testing Awareness     |
| ✓ Published Practitioner in AI Agent Testing        | ✓ Agile, TDD, BDD, Shift-Left Quality Engineering      |

## PROFESSIONAL EXPERIENCE

**Independent Development** | Fremont, CA (Remote)

August 2022 to Present

### **Senior SDET and AI Test Automation Architect**

- **AI Agent-Driven Exploratory Testing (Published):** Designed and implemented an AI agent system that performs exploratory testing against running applications, hunts for untested risks, and generates targeted test cases to close coverage gaps. Documented in published LinkedIn article. The agent uses the Anthropic Claude SDK (same tooling SFC uses via Claude Code) for exploratory reasoning, prompt engineering, and test generation.
- **LLM-Powered Test Generation at Scale:** Built Python/pytest/Playwright framework using Anthropic Claude SDK to auto-generate 10-20 targeted test cases per OpenAPI spec covering happy paths, adversarial edge cases, and coverage gaps. 145+ passing tests across 17 parallel GitHub Actions CI jobs. Full portfolio at [github.com/njmarshall/ai-test-automation-python](https://github.com/njmarshall/ai-test-automation-python).
- **DeepEval LLM-as-Judge Evaluation:** Implemented DeepEval evaluation layer scoring AI-generated test outputs for hallucination, coverage, and correctness before merge. Addresses SFC's requirement to make test results reliable and useful for engineers investigating bugs and preparing releases.
- **Self-Healing Orchestration with JIRA Audit Trail:** Evolved AST-parsing StalenessDetector into a self-healing loop that flags broken tests when API contracts change and drives regeneration. Extended with Phase 6 JIRA deviation audit trail requiring human sign-off on every healing event.
- **AI Observability and Drift Detection:** Instrumented AI call logging capturing prompts, responses, latency, and cost per LLM call with drift detection to surface model behavior changes over time. Directly applicable to monitoring agent-driven test system health.

- **Postgres and Data Validation:** SQL is my strongest and longest-held skill at 12+ years across PostgreSQL, MySQL, Oracle, and SQL Server. Complex JOIN queries, data integrity checks, and test data setup/teardown for comprehensive scenario coverage.

**Finix** | San Francisco, CA

March 2022 to July 2022

#### Software Engineer in Test

- **Payment System Quality Engineering:** Designed test strategies for PCI-regulated payment microservices covering REST API and event-driven billing flows. Cypress UI automation, Cucumber BDD, CircleCI CI/CD certification with AWS log validation.
- **Departure:** Position ended July 2022 as part of a company-wide reduction during the 2022-2023 tech industry correction, which eliminated 165,000+ tech workers in 2022 alone.

**Indeed** | San Francisco, CA

May 2018 to August 2021

#### QA Automation Engineer

- **Automated Test Systems at Scale:** Designed and owned Python/pytest/gRPC automated test framework for backend notification servers serving 250M+ monthly users. 100+ backend test scripts covering notification delivery pipelines processing millions of events daily with Jenkins CI/CD automation.
- **Making Test Results Reliable and Useful:** Diagnosed and resolved async race condition causing 30% test flakiness. Authored 30+ internal wiki pages and delivered 10 engineering presentations to help engineers investigate bugs and prepare releases effectively.

**HEAVY.AI (formerly MapD)** | San Francisco, CA

June 2017 to February 2018

#### Automation Engineer

- **Distributed System Test Automation:** Led backend test automation for distributed GPU analytical database using Google Test/C++, Kafka, AWS, and Node.js. Authored wiki guides enabling engineering team independence.
- **Departure:** Role scope shifted significantly post-hire; transitioned to pursue roles with stronger QA ownership and automation mandate.

**Dalet Digital Media (Ooyala)** | Santa Clara, CA

September 2013 to October 2016

#### SDET, Data Engineering

- **Large-Scale Automated Test Systems:** Built test automation frameworks for end-to-end data pipelines using Hadoop, Cassandra, Spark, Java, Go, Scala/BDD, Gatling, JMeter, Docker, and AWS. 750+ test cases, 90%+ coverage, 25+ Jenkins jobs.

**Triplt (Concur)** | San Francisco, CA

November 2011 to April 2013

#### Software Test Engineer

- **UI and API Automation:** Java/JUnit/Selenium framework with 80+ page objects, OAuth automation, REST API test suite. 750+ test cases, 150+ bugs logged.

**Salesforce** | San Francisco, CA

December 2009 to August 2011

#### Quality Engineer

- **Enterprise SaaS Quality:** 400+ bugs logged, 630+ test cases with 90%+ coverage across CRM cloud sync and ISVForce features.

**Microsoft** | Redmond, WA

February 2007 to June 2009

#### SDET and Web Developer

- **Full-Stack Background:** API, UI, and database test automation for embedded devices. JavaScript and HTML front-end development background relevant to TypeScript and React familiarity.

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## TECHNICAL SKILLS

**Languages:** Python (primary scripting), Java, JavaScript, TypeScript, Go, Scala, C#, SQL, Bash

**AI and LLM:** Anthropic Claude SDK, Claude Code, OpenAI GPT, DeepEval LLM-as-judge, AI agent exploratory testing, self-healing orchestration, AI observability, drift detection

**Test Frameworks:** pytest, Playwright, Selenium WebDriver, TestNG, JUnit 5, RestAssured, Cucumber/BDD, Cypress, Appium, JMeter, Gatling, k6

**Databases:** PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Cassandra, Apache Spark, Hadoop, JDBC

**CI/CD and DevOps:** GitHub Actions, Jenkins, CircleCI, Docker, Kubernetes, AWS, GCP (adjacent), Git, Gradle, Vercel (adjacent)

**API and Integration:** REST, GraphQL, gRPC, OpenAPI/Swagger, Pact contract testing, OAuth2/JWT, Postman

**Reporting and Tools:** Allure, TestRail, JIRA, Confluence, Sentry (adjacent), Linear (aware), Postman

**Methodologies:** Agile/Scrum, TDD, BDD, Shift-Left Testing, Agent-Driven Testing, Risk-Based Testing

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## EDUCATION AND CONTINUING DEVELOPMENT

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B.A., Computer Science and Mathematics (Double Major) | Gallaudet University, Washington, DC

AI Fundamentals: Think, Create, and Work Smarter | MasterClass, 2026

Postgraduate Program in Cloud Computing (AWS) | University of Texas at Austin, 2024

Scaler Academy | System Design, Backend APIs, Databases, Algorithms | 2022-2024